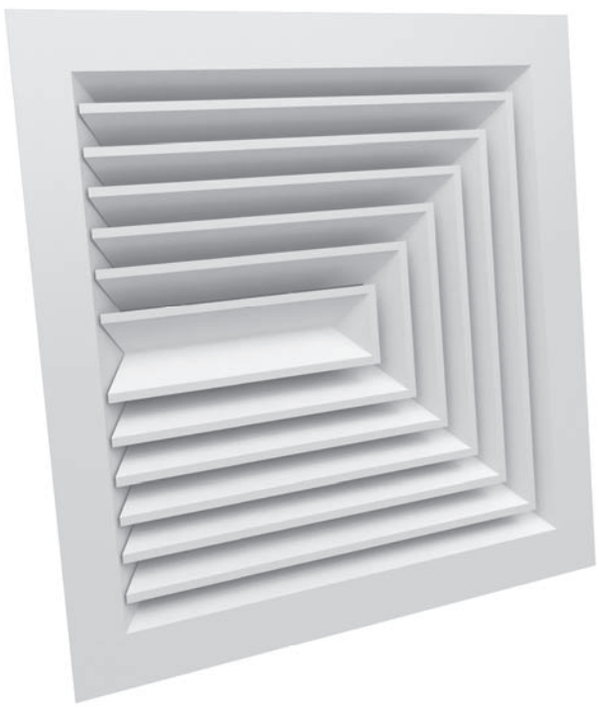
Ceiling Diffuser CD-S



CD-S3

Description
1, 2 or 3 direction square diffuser preferred for paneled ceiling mounting.

Application
Supply/ exhaust of airflow.

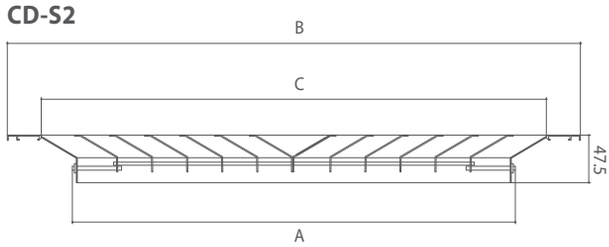
Characteristics
Recommended for rooms of a maximum height of 4 meters .
Removable central cone for easy installation and cleaning.
Suitable for horizontal air supply.
The difference between the recommended supply air temperature and the room air temperature is ± 10 K.
The air flow can be adjusted by mounting a air control damper **OBD** on the diffuser inlet.

Accessories
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

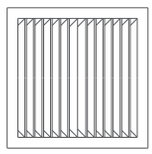
Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request.
Natural anodised aluminum.

Technical design

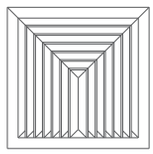
CD-S2



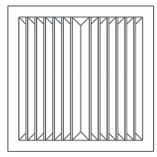
CD-S1



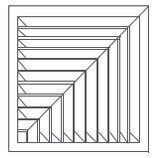
CD-S3



CD-S2



CD-S2-90



Assembly Methods
Paneled ceiling 600 x 600 mm (standard)
Mounting holes on diffuser connection (optional)

Ceiling Diffuser CD-S

Selection table CD-S

Air flow (m³/h)	LxH	150 x 150			225 x 225			300 x 300			375 x 375			455 x 455			525 x 525			600 x 600		
	Model	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3
	Ak(m²)	0.0096	0.0093	0.0088	0.0215	0.0209	0.02	0.0383	0.0373	0.0355	0.0598	0.0582	0.0554	0.0863	0.0838	0.0798	0.1174	0.1141	0.1086	0.1534	0.149	0.1418
100	Veff [m/s]	2.9	3.0																			
	X (m)	2.2	0.8																			
	NR [dB(A)]	22	22																			
	ΔPt(Pa)	5.9	6.7																			
160	Veff [m/s]	4.6	4.8	5.1																		
	X (m)	3.5	1.2	1.2																		
	NR [dB(A)]	33	33	29																		
	ΔPt(Pa)	15	17.1	22																		
200	Veff [m/s]	5.8	6.0	6.3	2.6	2.7	2.8															
	X (m)	4.4	1.5	1.5	2.9	1	1															
	NR [dB(A)]	39	39	34	22	22	<20															
	ΔPt(Pa)	23.4	26.8	34	4.7	5.3	7															
300	Veff [m/s]	8.7	9.0	9.5	3.9	4.0	4.2	2.2	2.2													
	X (m)	6.6	2.3	2.3	4.4	1.5	1.5	3.3	1.1													
	NR [dB(A)]	49	49	44	32	32	28	20	21													
	ΔPt(Pa)	52.7	60.2	77	10.5	11.9	15	3.3	3.7													
400	Veff [m/s]				5.2	5.3	5.6	2.9	3.0	3.1	1.9	1.9										
	X (m)				5.8	2	2	4.4	1.5	1.5	3.5	1.2										
	NR [dB(A)]				39	40	35	27	28	23	<20	<20										
	ΔPt(Pa)				18.7	21.2	27	5.9	6.7	9	2.4	2.7										
600	Veff [m/s]				7.8	8.0	8.3	4.4	4.5	4.7	2.8	2.9	3.0	1.9	2.0							
	X (m)				8.8	3	3	6.6	2.3	2.3	5.3	1.8	1.8	4.4	1.5							
	NR [dB(A)]				49	50	45	37	38	33	28	29	24	20	21							
	ΔPt(Pa)				42.1	47.7	61	13.3	15	19	5.4	6.2	8	2.6	3							
800	Veff [m/s]							5.8	6.0	6.3	3.7	3.8	4.0	2.6	2.7	2.8	1.9	1.9				
	X (m)							8.8	3	3	7	2.4	2.4	5.8	2	2	5	1.7				
	NR [dB(A)]							44	45	40	35	36	31	27	28	24	21	22				
	ΔPt(Pa)							23.6	26.6	34	9.7	10.9	14	4.6	5.3	7	2.5	2.8				
1000	Veff [m/s]							7.3			4.6	4.8	5.0	3.2	3.3	3.5	2.4	2.4	2.6	1.8	1.9	
	X (m)							11			8.8	3	3	7.3	2.5	2.5	6.3	2.2	2.1	5.5	1.9	
	NR [dB(A)]							50			41	41	37	33	34	29	26	28	23	21	22	
	ΔPt(Pa)							36.8			15.1	17.1	22	7.3	8.2	11	3.9	4.4	6	2.3	2.6	
1200	Veff [m/s]										5.6	5.7	6.0	3.9	4.0	4.2	2.8	2.9	3.1	2.2	2.2	2.4
	X (m)										10.5	3.7	3.6	8.8	3	3	7.5	2.6	2.6	6.6	2.3	2.3
	NR [dB(A)]										45	46	41	37	38	34	31	32	28	25	27	22
	ΔPt(Pa)										21.7	24.6	32	10.4	11.9	15	5.6	6.4	8	3.3	3.8	5
1600	Veff [m/s]										7.4		8.0	5.1	5.3	5.6	3.8	3.9	4.1	2.9	3.0	3.1
	X (m)										14		4.8	11.7	4.1	4	10	3.5	3.4	8.8	3	3
	NR [dB(A)]										52		48	44	46	41	38	39	35	33	34	29
	ΔPt(Pa)										38.7		56	18.6	21.2	27	10	11.4	15	5.9	6.7	9
2000	Veff [m/s]																	4.9	5.1		3.7	3.9
	X (m)																	4.3	4.3		3.8	3.8
	NR [dB(A)]																	45	40		39	35
	ΔPt(Pa)																	17.8	23		10.4	13
3000	Veff [m/s]																		7.7		5.6	5.9
	X (m)																		6.4		5.7	5.6
	NR [dB(A)]																		50		49	45
	ΔPt(Pa)																		51		23.5	30

NR < 35 | 35≤NR≤45 | NR>45

Annotation

* Ak [m²] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss
* Vk = 0.25 [m/s] - Speed at which air jet length was calculated "x" [m]

Standard dimensions

CD-S 1,2,3			
neck / external size	**A**	**B**	**C**
150 / 290	150 x 150	290 x 290	230 x 230
150 / 595	150 x 150	595 x 595	235 x 235
225 / 365	225 x 225	365 x 365	305 x 305
225 / 595	225 x 225	595 x 595	315 x 315
300 / 440	300 x 300	440 x 440	380 x 380
300 / 595	300 x 300	595 x 595	385 x 385
375 / 515	375 x 375	515 x 515	455 x 455
375 / 595	375 x 375	595 x 595	465 x 465
455 / 595	455 x 455	595 x 595	535 x 535
480 / 620	480 x 480	620 x 620	560 x 560
525 / 665	525 x 525	665 x 665	605 x 605
600 / 740	600 x 600	740 x 740	680 x 680

Order codes

Type	–	**No. of directions**	+	**OBD**	+	**ADP IZ/N**	/	**Dimensions**	/	**L X H**	/	**O**
CD-S		1, 2, 3		Air control damper		Insulated/ Non-insulated adapter		**neck / external** 150 / 290 150 / 595 225 / 365 225 / 595 300 / 440 300 / 595 375 / 515 375 / 595 455 / 595 480 / 620 525 / 665		Other dimensions on request		**0** Painted white RAL 9016 **R** Other RAL paints Natural anodised